PDF with Markdown data

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EXCLUSIVE INVESTIGATION

Here's how to generate PDFs in Python with dynamically generated diagrams

This example reads a Markdown-formatted data table and generates a PDF document including charts generated from that data.

BY THE HOME DISPATCH DATA DESK · CLIMATE CORRESPONDENT

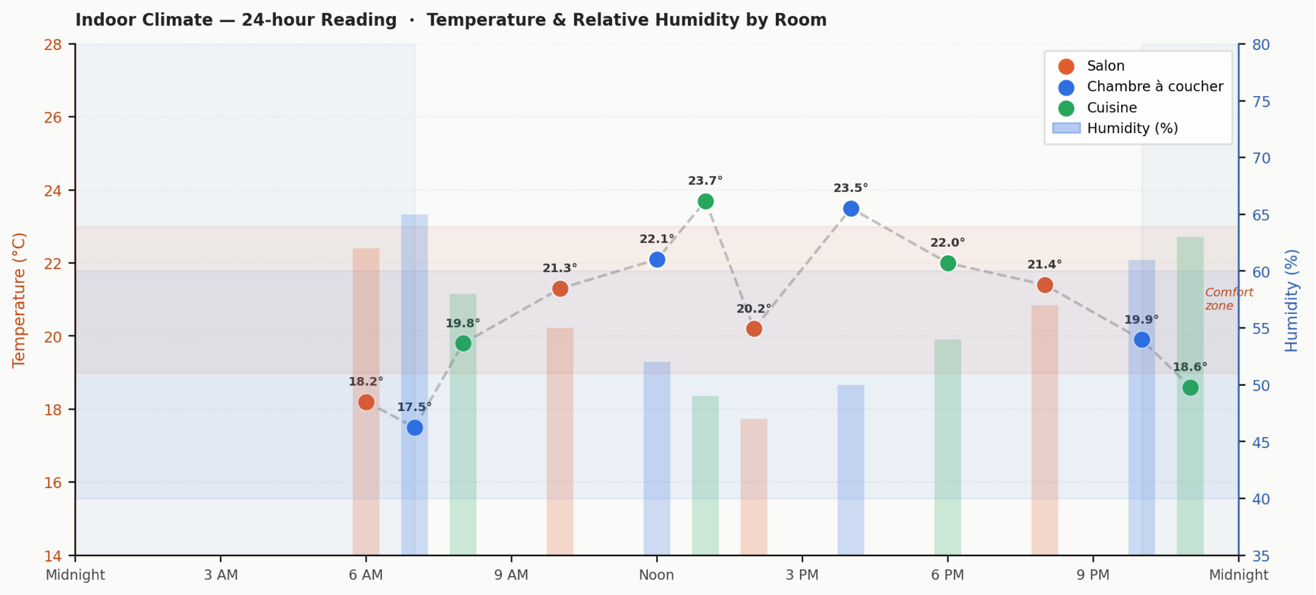

FIGURE 1 — Hourly temperature readings (coloured markers) against humidity bars across three rooms. The shaded red band marks the 19–23 °C comfort zone; blue background shading denotes overnight hours.

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MARTIGNY-VILLE, MONDAY — A full-day monitoring sweep of a residential property has produced a detailed climate portrait rarely seen outside professional buildings. Twelve discrete measurements, taken between 06:00 and 23:00 across three rooms, reveal a textbook daily temperature cycle — rising steadily from a morning low of 17.5 °C to an afternoon peak of 23.7 °C before dropping back toward overnight lows.

The average temperature across all readings stood at 20.7 °C, marginally above the widely cited 20 °C threshold for thermal comfort in temperate climates. Crucially, the warmest room at any given hour was the Kitchen, which benefits from heat produced by cooking appliances and solar gain through a south-facing window.

Meanwhile, the Bedroom consistently recorded the coolest temperatures — a phenomenon well-documented in the sleep-science literature, which recommends sleeping environments between 16 °C and 19 °C for optimal rest. Residents will be pleased to note that the Bedroom readings remain within or just above this range throughout the survey period.

HUMIDITY SWINGS TRACKED ACROSS ZONES

Relative humidity showed an inverse relationship to temperature, as expected from basic psychrometrics. At the morning low of 17.5 °C, humidity reached 65 %, falling to a midday minimum of 47 % as the air warmed and its capacity to hold moisture increased. The mean relative humidity of 56 % sits within the 40–60 % band recommended by the World Health Organization for indoor air quality.

> *"A well-ventilated home should show exactly this kind of inverse temperature-humidity rhythm throughout the day."*

Prolonged exposure to humidity above 65 % can encourage mould growth and dust-mite proliferation, two of the leading causes of respiratory irritation in domestic settings. The data suggest that, while early-morning readings edge toward this boundary in the Bedroom, the house recovers quickly once heating begins.

SENSORS & METHODOLOGY

All readings were obtained via calibrated DHT-22 sensors, accurate to ±0.5 °C and ±2 % RH. Timestamps reflect local civil time. Data were logged to a Raspberry Pi home server and cross-checked against a reference thermometer placed in each room for the duration of the survey.

Sensors were mounted at seated head-height (approximately 1.2 m) away from direct sunlight, radiators, and draughts. The Kitchen sensor was positioned above the worktop, clear of steam sources. No readings were discarded or adjusted.

Full raw data, including sensor calibration certificates and server logs, are available on request from the Home Automation Data Desk.

RAW READINGS AT A GLANCE

Time	Location	Temp (°C)	Humid. (%)
06:00	Salon	18.2	62
07:00	Chambre à coucher	17.5	65
08:00	Cuisine	19.8	58
10:00	Salon	21.3	55
12:00	Chambre à coucher	22.1	52
13:00	Cuisine	23.7	49
14:00	Salon	20.2	47
16:00	Chambre à coucher	23.5	50
18:00	Cuisine	22.0	54
20:00	Salon	21.4	57
22:00	Chambre à coucher	19.9	61
23:00	Cuisine	18.6	63

All 12 readings from the survey period, ordered by time of day.